

Hazards and incidents: Detection and learning in radiation medicine, a comparison of 2 educational interventions

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Abstract

Purpose: Interprofessional, educational live simulations were compared with group discussion-based exercises in terms of their ability to improve radiation medicine trainees' ability to detect hazards and incidents and understand behaviors that may prevent them.

Methods and materials: Trainees and recent graduates of radiation therapy, medical physics, and radiation oncology programs were recruited and randomized to either a simulation-based or group discussion-based training intervention. Participants engaged in hazard and incident detection, analysis, and a discussion of potential preventive measures and the concept of the "highly reliable team." A video examination tool modeled on actual incidents, using 5-minute videos created by faculty, students, and volunteers, was created to test hazard and incident recognition ability before

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and after training. Hazard and incident detection sensitivity and specificity analyses were conducted, and a survey of the participants' and facilitators' perceptions was conducted.

Results: Twenty-seven participants were assigned to the simulation ($n = 15$) or discussion group ($n = 12$). Hazard and incident-detection sensitivity ranged from 0.04 to 0.56 before and 0.04 to 0.35 after training for the discussion and simulation groups, respectively. The pre- and posttraining difference in sensitivity between groups was 0.03 ($P = .75$) for the minimum and 0.33 ($P = .034$) for the maximum reaction time. Participant perceptions of the training's educational value in a variety of domains ranged from a mean score of 6.58 to 8.17 and 7 to 8.07 for the discussion and simulation groups, respectively. Differences were not statistically significant. Twenty-six of the 27 participants indicated that they would recommend this event to a colleague.

Conclusions: Participants' ability to detect hazards and incidents as portrayed in 5-minute videos in this study was low both before and after training, and simulation-based training was not superior to discussion-based training. However, levels of satisfaction and perceptions of the training's educational value were high, especially with simulation-based training.

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Introduction

Despite high quality standards, incidents still occur in modern radiation medicine programs and have the potential to harm patients.¹ An "incident" may be defined as "an unwanted or unexpected change from a normal system behavior which causes or has the potential to cause an adverse effect on persons or equipment."² Hazards, or risks, may increase the likelihood of an incident. Reported rates of radiation therapy incidents range between 0.2% and 2% per course; fortunately, the vast majority of incidents have either no or minimal clinical impact.^{3,4} However, these incident rates are higher than those in anesthesia, transfusion medicine, or the airline industry.⁵⁻¹⁰ Published trend analyses report that common causes of radiation therapy incidents include poor communication and errors in transmission of essential information¹¹⁻¹³; therefore, in addition to the technical aspects of treatment, quality assurance in radiation medicine should also address all of the interdisciplinary interactions that occur along the patient's path from initial consultation to posttreatment follow-up.¹⁴ Team-based interactions are difficult to change and manage, however.

We provided an interprofessional educational course at our institution that aimed to improve radiation medicine trainees' ability to detect hazards and incidents and understand behaviors that can prevent them. Although simulation, compared with no intervention, has been shown to improve the knowledge, skills, and behaviors of health professionals, it remains to be determined whether simulation is superior to less resource-intensive educational interventions.¹⁵ The primary aim of this study therefore was to compare a live simulation-based exercise with a group discussion in terms of improving trainees' hazard and incident-detection ability and their understanding of preventive behaviors. The secondary aim of the study was to compare participants' perceptions of these 2 interventions.

Materials and methods

Institutional research ethics board approval was obtained. Participants were recruited from the University of Toronto and the Michener Institute programs in radiation therapy. A 3-year degree after 1 or more years of postsecondary education; radiation oncology, a 5-year residency after a doctor of medicine; and medical physics, a 2-year residency after a graduate degree. All of the activities described in this report occurred at the Michener Institute campus building with meeting rooms with audiovisual equipment and external beam radiation therapy and imaging simulation suites that included full-size training linear accelerators and computed tomography scanners.

Simulation group and discussion group

Participants were stratified by discipline and were randomly assigned to either a simulation or discussion group. Each exercise was approximately 2.5 hours in length and was composed of 3 groups of 4 or 5 participants who rotated through 3 stations (Table 1), each taking about 45 minutes and led by a faculty member facilitator. The majority of facilitators, in addition to their education role, also had a career focus on quality control and assurance. The objectives of these training sessions were to examine and practice detecting previous actual hazards and incidents, learn about potential preventive measures, and learn about the concept and practical aspects of a highly reliable team.¹⁶ Before these exercises, all participants attended a didactic lecture that introduced the Canadian Partnership for Quality Radiotherapy and its National System for Incident Reporting in Radiation Treatment taxonomy¹⁷ and reviewed the frequency and types of radiation therapy incidents worldwide.

In the simulation stations, participants initially observed a team of faculty volunteers and simulated patients demonstrating, through acting, a scenario that contained at least 1 hazard or incident. Trainees were encouraged to

Table 1 Summary of content of training stations

Station	Summary of content of training station	Additional discussion points
1. Radiation therapy for spinal cord compression	Incorrect treatment setup position and on-board image guidance acquisition, incorrect segment of spinal column treated	Definition of medical event, disclosure of incident to patient and how to disclose
2. Intensity modulated radiation therapy for oropharynx carcinoma	Software and hardware error, leading to treatment delivery with open (uncollimated) fields and severe toxicity	Communication strategies with peers when there is a difference in opinion, ensuring compliance with quality assurance procedures
3. Adjuvant radiation therapy for early-stage breast cancer	Patient with multiple breast scars who received a boost treatment to the incorrect location because of incorrect target definition	Diffusing a power differential, creating a safety culture that encourages speaking up

interject, ask questions, and point out any hazards or incidents as they occurred. If the trainees did not detect a hazard or incident, the scenario was left to proceed to 1 of several consequences leading to potential patient harm, at which point it was stopped by the facilitator. Subsequently, each trainee “played” the roles of their respective discipline team members in the scenario and repeated the sequence of the events themselves. Finally, a facilitator-led debrief was conducted in which participants were asked to discuss the details of the incident in terms of the initiating factors and potential preventive measures using concepts of a highly reliable team. Similarly, participants randomized to the discussion sessions were presented with a script that described the scenes in the live simulation. This was read as a group, and then participants engaged in a discussion, led by a facilitator, similar to the previous postsimulation discussion. In preparation for the training event, facilitators engaged in a 2-hour general training session and a 30-minute tutorial specific to their station and were provided with a standardized written discussion guide for their station before the event.

Pre- and posttraining video evaluation tools

A series of 12 videos (approximately 5 minutes each) was produced to measure participants’ ability to detect radiation therapy hazards and incidents before and after the training activity. Eight videos, taken from our clinical incidents database, were designed to portray a specific hazard or incident. Four videos did not contain any hazard or incident; they showed normal day-to-day practice. The videos were viewed through a custom-made Web-based interface that allowed the users to view the video and click a button at the exact moment that they believed a hazard or incident occurred; the time (minutes, seconds) in the video at which the button was clicked, or the time stamp, was recorded (see Fig E1; available as supplementary material online only at www.practicalradonc.org). A panel of 4

faculty members with extensive experience in incident reporting and assessment reviewed all 12 videos. Although no formal interrater reliability analysis was performed, 10 videos were selected (Table E1) for the study based on a high degree of consensus for the presence or absence of a hazard or incident.

One week before the training session, all participants were required to view the first group of 5 videos using the Web interface. Participants were given unique number identifiers to log into the Web interface and view the videos sequentially. They were instructed to click a button in the Web interface if they detected a hazard or incident while watching the video. The time stamp was recorded. The time stamps were compared against the consensus times obtained from the 4 faculty members to contrast the performance of trainees to that of staff with established expertise in incident analysis. One week after the educational event, the participants were required to complete another 5 video evaluations that were different from the pretraining videos (Fig 1).

Participant perceptions

All participants were asked to complete a questionnaire immediately before and after the training event; faculty facilitators were asked to complete a postevent questionnaire. Responses to questions, summarized in the Results section, were ranked according to the Likert scale, or were free-text responses.

Statistical analysis

In videos that contained an identifiable hazard or incident, there were 4 possible responses: either viewers clicked before, during, or after the hazard or incident (r_1 , r_2 , r_3 , respectively) or they do not click at all (r_4) (Fig 2). The start of the hazard or incident, t_0 , was defined as the minimum click time of the expert reviewers minus 5 seconds; t_x was defined as the possible detection time delay or reaction time.

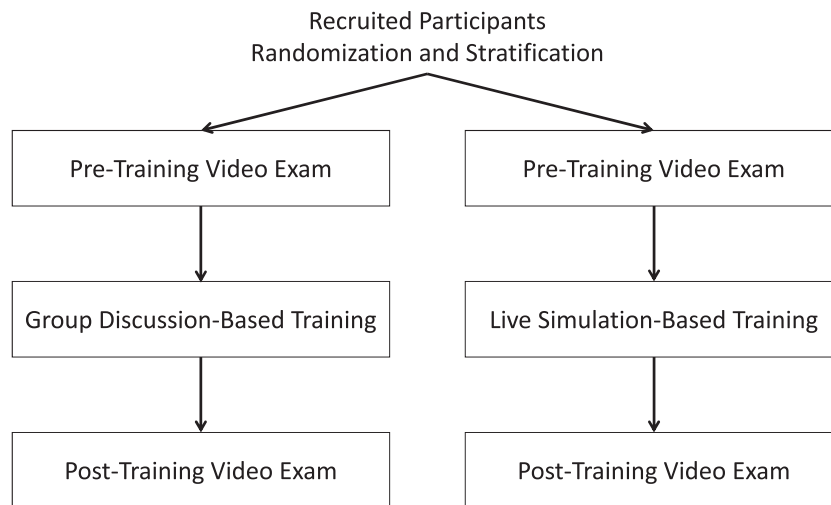


Figure 1 Overall study schematic.

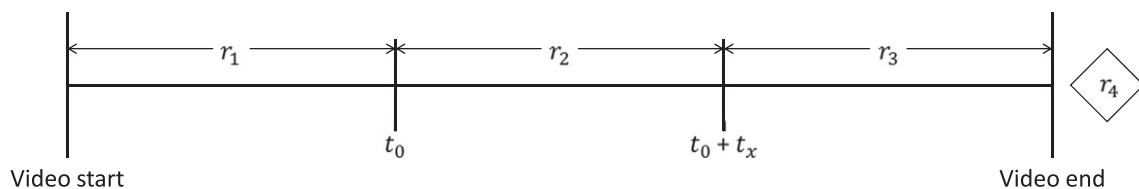


Figure 2 Categorization of “click times” for videos with a hazard or incident. r_1 , r_2 , r_3 = possible time intervals during which a viewer may detect an incident and click; r_4 = viewer does not click at all; t_0 = the minimum click time of the expert reviewers minus 5 seconds; t_x = the possible reaction time delay to detect an incident.

Sensitivity was calculated using

$$\text{Sensitivity} = \frac{\#r_2}{\#r_1 + \#r_2 + \#r_3 + \#r_4}$$

To perform a sensitivity analysis and to avoid biases from an arbitrarily set value of t_x , sensitivity was calculated using t_x values of varying length for each video. Similarly, in the case of videos for which it was determined that there was no identifiable hazard or incident, there were 2 possible responses: s_1 , the viewer incorrectly clicked at any point during the video; or s_2 , the viewer did not click at all. Specificity was calculated using

$$\text{Specificity} = \frac{\#s_2}{\#s_1 + \#s_2}$$

The Mann-Whitney test was used to determine if the changes in sensitivity and specificity before and after the training event differed between the simulation and discussion groups. For the questionnaires, the Wilcoxon signed rank test was used to compare paired data (ie, pre- and posttraining event Likert-scale scores). The Mann-Whitney test was used to compare scores and changes in scores between groups.

Results

Participant characteristics

Thirty participants were enrolled in the study; however, 3 did not attend the training session. Participant characteristics are summarized in Table 2.

Sensitivity and specificity analysis

Of the 27 participants included in the analysis, 5 did not complete the video evaluations and were therefore

Table 2 Participant characteristics

	Simulation group (n = 15)	Discussion group (n = 12)
Gender		
Female	6	4
Male	9	8
Profession		
Radiation oncology	3	3
Medical physics	4	5
Radiation therapy	8	4
Median clinical experience	2 y	2 y

Table 3 Sensitivity Analysis for pre- and posttraining video examinations

t_x (s)	Group	Pretraining		Posttraining		Median pre- vs. posttraining difference (Q1-Q3)	Discussion vs. simulation difference <i>P</i> value
		Mean sensitivity	Median sensitivity	Mean sensitivity	Median sensitivity		
5	Discussion	0.06	0	0.04	0	0 (0-0)	.75
	Simulation	0.04	0	0.05	0	0 (0-0)	
15	Discussion	0.22	0	0.09	0	0 (-0.5 to 0)	.24
	Simulation	0.04	0	0.09	0	0 (0-0)	
30	Discussion	0.28	0.5	0.13	0	0 (-0.5 to 0)	.099
	Simulation	0.08	0	0.17	0	0 (0-0.33)	
60	Discussion	0.44	0.5	0.22	0	0 (-0.5 to 0)	.068
	Simulation	0.15	0	0.22	0.33	0 (0-0.17)	
90	Discussion	0.56	0.5	0.31	0.5	-0.33 (-0.5 to 0)	.034
	Simulation	0.27	0	0.35	0.33	0 (-0.17 to 0.33)	

Q1-Q3 = first to third quartiles; t_x = possible reaction delay after the expert detected an incident.

excluded from the sensitivity and specificity analyses. Two videos (Table E1) were converted from videos that did contain a hazard or incident to videos that did not because of very low participant hazard or incident detection. Sensitivity estimates were calculated for a variety of t_x shown in Table 3. Specificity did not differ significantly between the 2 groups.

Participant perceptions

Responses to the question “To what degree have you interacted with the other radiation medicine professionals in your time in the training program?” revealed that of the participants from radiation therapy, 40% and 50% never or very rarely interact with radiation oncologists and medical physicists, respectively. Similarly, 50% and 22% of medical physics trainees replied that they never or very rarely interacted with oncologists or therapists, whereas 0% and 33% of radiation oncology trainees felt that they never or very rarely interacted with therapists and medical physicists, respectively.

The level of clinical scenario complexity was judged to be “just right” by 78%, 83%, and 82% of medical physics, radiation oncology, and radiation therapy participants, respectively. Two medical physics participants considered the scenarios too advanced but there were no radiation oncology or therapy participants who considered them too advanced. All 6 of the facilitators felt that the clinical scenarios’ level of complexity was appropriate for the participants’ level of experience.

Participants’ and facilitators’ perceptions of the value of the training event, after they had completed it, are shown in Table 4. Differences between the 2 participant groups were not statistically significant.

Twenty-six of 27 postevent survey respondents stated that they would recommend this event to another colleague.

Discussion

The results stated in the Participant perceptions section suggest that there is a need to improve team-based learning opportunities at our institution. In an era of competency-based training and evaluations, such as that promoted and adopted by the Royal College of Physicians and Surgeons of Canada,¹⁸ these opportunities for practical interprofessional collaboration are essential. Simulation has been used in a variety of fields of medicine including surgery, anesthesia, and emergency medicine,¹⁹⁻²⁴ but has yet to make a significant impact in the field of radiation oncology. To our knowledge, this is the first study comparing 2 different approaches to interprofessional learning in radiation medicine: simulation, a resource-intensive intervention, and discussion-based training, which may be less costly.

The sensitivity data presented, as determined by the calculation provided previously and in Fig 3, indicated that in both groups participants’ ability to detect an incident or hazard in the video examinations was less than or slightly better than a result expected by chance alone, or a sensitivity of 0 to approximately 0.33, depending on t_x and t_0 . This suggests that the video examination method did not capture the participants’ ability to detect hazards and incidents well, possibly because the portrayed incidents were too subtle or not clearly depicted, as suggested by the fact that 2 videos originally intended to contain a hazard or incident were reclassified as videos without one. A comparison of sensitivity data from the simulation and discussion groups suggests those in the simulation group showed a greater improvement in detection ability when t_x was set to 30, 60, and 90 seconds, although statistical significance was only reached for $t_x = 90$ seconds. This was likely because of imbalance in pretraining sensitivity between the groups, with the discussion group showing superior pretraining detection sensitivity.

Table 4 Posttraining event survey responses

Question	Participants			Facilitators' perceptions of event's impact on participant	
	Discussion group mean score ^a (SD)	Simulation group mean score ^a (SD)	Simulation minus discussion group	Discussion group mean score ^a	Simulation group mean score ^a
To what extent did this event improve your:					
Ability to recognize and detect radiation therapy incidents	6.58 (1.38)	7.33 (1.5)	+0.75 NS	10.0	5.7
Knowledge of factors that can contribute to a radiation therapy incident	7.42 (1.38)	8.07 (1.49)	+0.65 NS	10.0	7.7
Knowledge of quality assurance components or factors that can prevent a radiation therapy incident	7.33 (1.23)	7.53 (1.68)	+0.2 NS	10.0	8.0
Ability to prevent future incidents	7 (0.6)	7.67 (1.5)	+0.67 NS	10.0	4.7
Familiarity with the Canadian Partnership for Quality Radiotherapy and its incident Taxonomy and Reporting Agenda	6.73 (1.62)	7 (2.32)	+0.27 NS	6.3	3.7
Knowledge of a highly reliable radiation medicine team	7.25 (1.36)	7.33 (1.95)	+0.08 NS	6.3	6.7
Exposure to other trainees within radiation medicine	8.17 (0.83)	8.13 (1.41)	-0.04 NS	9.7	7.3
Interprofessional communication	7.83 (1.53)	7.93 (1.79)	+0.1 NS	9.7	7.0
Assertiveness and willingness to speak up when you recognize a hazard or incident	7.42 (2.11)	7.93 (1.16)	+0.51 NS	7.7	7.3
Overall clinical knowledge and skills	7.08 (1.38)	7.73 (1.67)	+0.65 NS	8.0	3.0
Do you think that this experience will shape or influence your future practice in radiation medicine?	7.36 (1.03)	8.4 (1.4)	+1.04 NS	NA	NA
How valuable was this event in your overall education in radiation medicine?	7.18 (1.33)	8.6 (1.3)	+1.42 NS	8.3	7.3
How important do you believe it would be for this event to be made an annual event for radiation medicine trainees at this institution?	7.54 (1.86)	9.07 (1.16)	+1.53 NS	8.3	9.0
Please rate your overall satisfaction with this event	7.73 (1.01)	9 (1)	+1.27 NS	7.7	9.0

NA = not applicable; NS = not statistically significant; SD = standard deviation.

^a Likert scale score: 1 = not at all, 10 = very much/extremely.

Despite the negative results from the sensitivity analysis, almost all participants and all facilitators anonymously indicated that they would recommend this event, which was voluntary, to a colleague. Although participant levels of satisfaction and perceptions of the event's overall value were high in both groups, they were highest in the simulation group, albeit not statistically significant. The small number of participants limited the statistical power of this study, but participant voluntary recruitment was probably as high as possible in a large academic radiation medicine training center. Another limitation of this study was the heterogeneity of participants both within the same profession and in different professions. This heterogeneity may have overwhelmed any improvement in detection ability brought about by either training intervention. Presumably, the educational

needs of each participant depend on their profession, experience, and education. Although the facilitators attempted to apply general teaching principles to each of the 3 professions, the teaching principles of each scenario could have been specifically tailored for each profession.

In addition to these results, participants and facilitators were asked to provide specific feedback and suggestions on how the event could be improved. Several participants stated that they appreciated the opportunity to discuss and practice the approach to disclosing an incident to a patient as well as how and when to issue an apology to a patient. Some suggested that the initial lecture could be made into a video that could be viewed before coming to the event to maximize the amount of time spent on interprofessional interaction. Participants in the discussion group felt that the case background could have been read in advance of

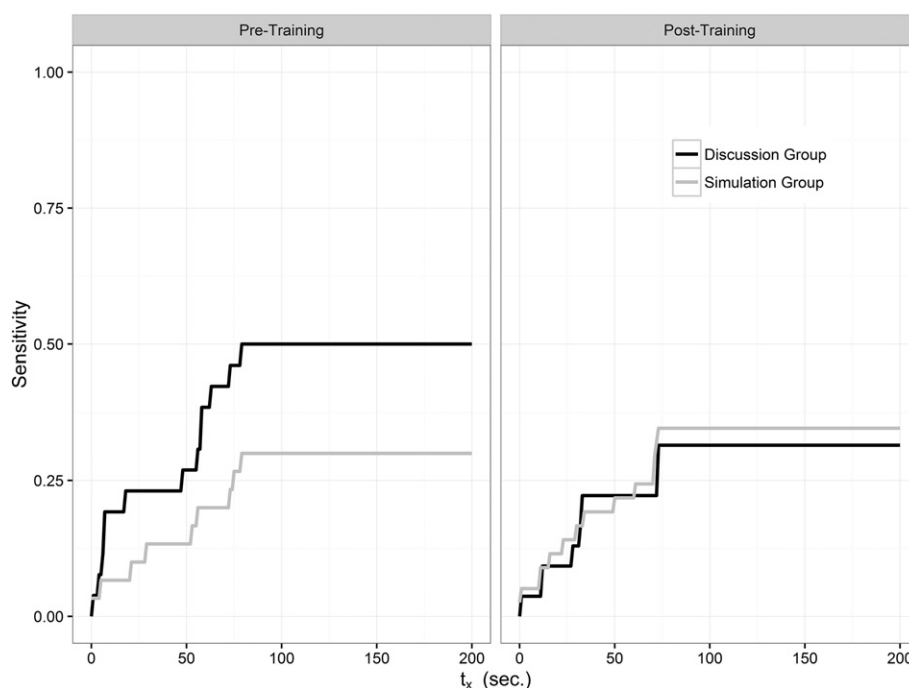


Figure 3 Pre- and posttraining sensitivity analysis. sec = seconds; t_x = possible reaction delay after the expert reviewer detected a hazard or incident.

the training session because they found reading the rather complex background information to be time-consuming.

In conclusion, live simulation, in comparison to discussion-based training, may be a more valued and appreciated teaching method for radiation medicine trainees. However, a video examination tool, as it was used in this study, did not demonstrate that simulation-based training was superior to discussion-based training in improving training hazard and incident detection ability.

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